

SEDAR 74 Gulf of Mexico Red snapper

LIFE HISTORY RESEARCH RECOMMENDATIONS

Age

- Resources are needed for personnel and database infrastructure to manage large, multidecade life history datasets that are beginning to exceed the capabilities of standard computers.
- Create a data repository with an upload interface for data providers to submit data directly into the SEDAR template. Build in standardized QA/QC methods for all data providers so that erroneous data points and outliers are identified and corrected prior to data workshops.
- Resume annual ageing workshops with gulf state agencies and other age data contributors to maintain high-quality age data given standard turnover rates among primary agers.
- Expand routine biological sampling, particularly in the eastern GOM subregion, where sample sizes are much lower compared to other subregions.
- The current subsampling protocol for red snapper is based on 5-year average landings by grid and is laborious and time consuming. Evaluate the current otolith subsampling protocol and provide alternatives to streamline the process.
- Evaluate the sampling design for observer programs.
- Investigate new technologies for estimating life history parameters (e.g. FT-NIRS, epigenetics) to increase production ageing efficiency and precision of age estimates.
- Increase sampling of sublegal fish through fishery independent surveys and the shrimp observer program to better estimate maturity and fecundity of smaller individuals, as well as samples through tournament intercepts to better estimate batch fecundity of larger/older females.

Growth

- Use inverse weighted age data for fitting growth curves.
- Estimate growth separately for each subregion with data from all years combined.

Reproduction

- Standardize data fields on the template, as well as limiting them to the data needed. It is especially important that data providers QA/QC their own data prior to submitting to ensure multiple fields are not used for the same parameter.

- Additional histological sampling is needed from the east region (FL west coast to Cape San Blas) to allow analyses by three regions.
- Adopt the slightly modified reproductive phase names and criteria from Lowerre-Barbieri et al. (2022).
- Adopt the standardized methodology from Lowerre-Barbieri et al. (2022) to estimate spawning season and peak spawning months.
- Maturity models should only use immature and spawning females (i.e., those with spawning markers) if sample size allows, rather than filtering data by peak spawning season, as recommended in Lowerre-Barbieri et al. (2022).
- Conduct batch fecundity estimates only on females in late oocyte maturation without POFs (histological analysis of ovaries used for batch fecundity is needed). Preserve ovaries only in formalin rather than Gilson's or freezing them. Use the washing process presented in Lowerre-Barbieri et al. (1993) for separating out the OM oocytes for fecundity estimates, which works equally well for fresh or preserved ovaries.
- Research on Red Snapper spawning marker duration, as well as selectivity of fish with spawning markers is needed to improve estimates of spawning frequency.
- Given the uncertainty in the fecundity-at-age vectors over time, utilize SSB as the measure of reproductive potential (Lowerre-Barbieri and Friess, 2022).

Natural Mortality

- Use the observed maximum age of 57 years when estimating age-specific M.
- Estimate a single M value and age-specific vector for all regions.
- Use the Then et al. (2015) method to estimate M using Lutjanid-specific parameters.
- Scale Then et al. (2015) derived estimate of M to age-specific values using Lorenzen function (1996).
- While important questions remain about density dependent effects on juvenile red snapper mortality, no new studies of age-0 and age-1 red snapper natural mortality were identified. All of the identified existing studies were considered in previous assessments, and their results are in line with the natural mortality rates for age-0 and age-1 red snapper used in SEDAR 31 and 52. Therefore, we recommend using $M = 2.0$ for age-0 and $M = 1.2$ for age-1 red snapper.
- We recommend additional effort to collect age-0 and age-1 red snapper to better estimate natural mortality rates and density dependent responses.

Episodic Events

- Further research is needed on the effects of episodic events on all life stages of red snapper.

COMMERCIAL FISHERY STATISTICS RESEARCH RECOMMENDATIONS

- Explore estimating gear selectivity using kept and discarded size data from the reef fish observer program.
- Investigate improving biological sampling of observer program by expanding sampling of otoliths paired with length data. Sampling should be completed without affecting fishing behavior and that may be possible by having sampling occur during breaks in fishing activity.
- Consider issuing research permits to fishers to retain catch below minimum size to collect samples for age length keys.
- Observer sampling may be supplemented by buying a percentage of catch for fish that cannot be extracted without causing damage to fish.
- Investigate trip ticket data for market category compared to length compositions. This analysis may provide some signal of age classes within the data.
- Consideration of the effect that resolutions of market category on trip tickets differ among states.

RECREATIONAL FISHERY STATISTICS RESEARCH RECOMMENDATIONS **Evaluation and Progress of Research Recommendations from Previous Assessments** Research recommendations from SEDAR 31 in 2013 were evaluated and progress on each item is outlined below:

1. Evaluate the technique used to apply sample weights to landings. Investigate the SEFSC method by analyzing the order of variables in the hierarchy and the minimum number of fish used. Furthermore, evaluate alternative methods, including a meta-analysis of the existing information from different sources, areas, states, surveys, etc. that could be performed.

Evaluation of Progress

- Clarity has been requested regarding the first line of this research recommendation. The sample weights here are referring to the weight of the fish sampled in APAIS and how those are used to calculate average weights for landings estimates in pounds whole weight. They do not refer to survey design sample weights used by MRIP to estimate catch.
 - The minimum number of fish used was evaluated in 2019 and an adjusted minimum sample size of 15 fish per strata was recommended and has been used since (SEDAR 67-WP-06).
 - Additional size information from LA BIO has been incorporated into the SEFSC weight estimation method since 2021.
2. Develop methods to identify angler preference and targeted effort. Require a reef fish stamp for anglers targeting reef fish, pelagic stamp for migratory species, and deep-water complex stamp

for deep-water species. The program would be similar to the federal duck stamp required of hunters and could help managers identify what anglers were fishing for.

Evaluation of Progress

- Florida requires private boat anglers to possess a State Reef Fish designation to legally possess a suite of reef fishes, including Red Snapper. This serves as a directory that is used to directly survey participants and estimate reef fish effort in Florida.

3. Continue and expand fishery-dependent at-sea observer surveys to collect discard information. This would help to validate self-reported headboat discard rates.

Evaluation of Progress

- Additional at-sea sampling programs for for-hire vessels have begun in Mississippi and Alabama and are described above in 4.6.2.1.
- The State of Florida dedicated recurring funds starting in 2020 to support this work long-term and provide stability. Data are available upon request for NOAA Fisheries to validate headboat discard rates.

4. Track Texas commercial and recreational discards.

Evaluation of Progress

- No progress noted

5. Estimate variances associated with the headboat program.

Evaluation of Progress

- Method developed in SEDAR 68 Research Track assessment for Scamp and described in SEDAR 68-DW-31.
- Alternative method described above in section 4.3.4 and recommended for use in SEDAR 74.

6. Evaluate existing and new methods to estimate historical landings. Hind-casting of Red Snapper landings is complicated by a lack of reliable historical effort data. To get at estimating historical effort, analysts could track consumables (gas, ice, bait) to develop price indices.

Evaluation of Progress

- No progress noted

7. Investigate how CPUE changes over time due to technological advances and changes in fishing practices.

Evaluation of Progress

- Adjusted ratios to account for technological advances from 1955 to 1980. These are described above in 4.4.5.
- Expanded years used in CPUE calculation to include 1981 to 1989, a period of time when the Red Snapper fishery was generally unregulated.

Research Recommendations for SEDAR 74

1. SSC to add TOR to operational assessment to include topical working group to review and evaluate the results of the Gulf of Mexico transition plan to optimize the use of state and federal data.
2. Integrate TPWD into the Gulf Transition Team in order to further evaluate the proposed calibration between TPWD and MRIP units and identify alternative methods that may be implemented, including increased benchmarking (e.g. 3-year benchmark period).
3. Gulf Transition Team should investigate the drivers of high MRIP wave specific effort estimates for recreational modes during traditionally low effort waves (e.g. winter waves, particularly in MS).
4. Develop and implement methods in the western Gulf region to collect vital statistics on the size distribution of recreational discards and directly estimate the magnitude of recreational discards in Texas.
5. Investigate the need for weighting headboat discard length composition data from new data streams. Determine if data need to be weighted due to over or under sampling of any particular trip types. If so, provide total number of trips sampled by state (or headboat region) and year, dock to dock hours for each trip, fleet (charter vs headboat), and catch type (harvest vs discard).
6. Investigate methods for weighting charter discard length composition data (to account for uneven sampling of trip types), or determine if weighting by trip type is necessary for that fleet.
7. Develop methods to properly weight discard length composition data from different states relative to the proportional magnitude of discards. 6 SEDAR 74 SAR SECTION V RESEARCH RECOMMENDATIONS January 2024 Gulf of Mexico Red Snapper
8. Develop statistically valid methods to identify outlier estimates (e.g. extremely high catches) and adjust sample weights for records that have a disproportionately high influence on total catch estimates, and establish new SEDAR best practice methods.
9. Provide working paper or presentations during the data workshop group meeting documenting collection methods and caveats for new data streams being evaluated / used.
10. Develop a list of qualitative information about the snapper-grouper fishery from stakeholders and methods to evaluate validity.
11. Research of additional reference points for historical landings.

12. Estimate and publish historical landings for major species (or species groups) in a single initiative to ensure a consistent methodology.
13. General evaluation of start year of existing models and value of historical data.
14. Evaluate how changes in fishing outcomes (fish for freezer vs. offshore experience with a few filets for dinner) have impacted fishing behavior over time. Important for determining validity of some historical landings assumptions.

MEASURES OF POPULATION ABUNDANCE RESEARCH RECOMMENDATIONS

- Explore alternative methods for properly weighting the DISL BLL survey in order to incorporate it with the NOAA Fisheries BLL survey
- Explore utility of design based index estimator for Gulfwide video survey
- Calibration of optical and acoustic imaging systems to better sample low visibility environments
- Explore alternative trip selection protocols that can account for changing regulations and angler behavior
- Explore influence of interacting species on gear selectivity and catchability

ASSESSMENT PROCESS RESEARCH RECOMMENDATIONS

Recommendations for considerations of future research are provided below in no particular order of priority. Recreational

Landings and Discards data

- Further develop best practices for correcting for prominent peaks and troughs in the earlier part of the time series where uncertainty is high and catch/discard estimates are driven by few but influential intercept records.
- Investigate influence of depredation as a contributor to discard mortality and its significance on observed discard data used in the assessment.

Composition Data Alternatives

- Incorporating age composition and length composition data for the directed fleets and estimating growth internally to the model to facilitate fit of multiple simultaneous sources of composition data.
- Consider the application of conditional age-at-length data for use in red snapper stock assessment.

Alternate Start Years

- SEDAR 74 moved the model start year from 1872 to 1950, but other later years would have been considered if not for modeling limitations. The determining factor in selecting 1950 was the

shrimp bycatch data and the lack of an ability to specify an initial F for a bycatch only fleet. This issue should be further explored and possible modifications to SS should be considered to allow the consideration of later start years.

Additional Data Needs

- Currently the model includes length-converted age composition data for surveys, where possible. It would benefit the model to include real age composition for trawl surveys in the future.
- Incorporating recreational discard composition into the east assessment area.
- Investigate the impact of using state survey derived landing statistics on the assessment model.

REVIEW PANEL RESEARCH RECOMMENDATIONS

Provide and Comment on Recommendations to Improve the Assessment Consider Data and Assessment Processes Research Recommendations, Improvement to the Assessment, and Additional Longer Term Research Recommendations

The Review Panel developed a number of conclusions about the assessment, and made recommendations about how some of the issues identified in the conclusions could be addressed. This section of the report shares our recommendations for data and model improvement in light of the conclusions reached about the assessment. The analysts followed the recommendations from the data and assessment meeting as discussed in the assessment report. Some examples of these included stock identification group three stock area model spatial partition, fixing natural mortality for age-0 and age-1 fish at values used during the last assessment, and trialing alternate start years for the model. Please note, however, that some of the recommendations in the stock assessment report are counter to the recommendations made by the Review Panel.

Recommendations for improvement or addressing inadequacies identified in the data or modeling

As discussed in the data section above, some of the data for model development were not adequately prepared for inclusion into the assessment model. The age and length composition data included in the model were raw and unscaled to the total landings or the distribution of the fleet or survey from which they were collected. The review panel recommended that composition data be properly scaled, as is often done in other SEDAR assessments in the Gulf of Mexico so that the model could be properly evaluated. For example, in a stratified random design, length samples from a fishery would need to be scaled up to the number of fish landed in each stratum to ensure that the length frequency is representative of the total landings from the fishery.

Both age and length data were included in the SEDAR 74 model. Generally speaking, age frequency data provides more information than an associated length frequency. Given that Stock Synthesis is an age-based assessment model, it is preferable to include age frequency data whenever possible. Given that the analysts reported that they had regional, annual, age-length keys generated using data from otolith analyses, we recommend that length frequency data be scaled, converted to age frequency data, and combined with any available scaled age data by year

and fleet for inclusion in the model. When age frequency data cannot be included in the model, then we recommend using length frequency if that is the only composition data available.

The configuration of the SEDAR 74 model that was presented to the reviewers was very different from the configuration used in the prior red snapper assessment. This was to be expected especially given the nature of a research track assessment, which is meant to explore alternative ways of configuring the assessment model and parsing the data. However, regarding the process, we recommend that continuity runs and bridging analyses be incorporated into the process. In this context, we are referring to continuity runs as those where the previous version of the model is updated with new data and run. Bridging is defined as making one change to the old model at a time and rerunning the model after each successive update, as an iterative methodology for arriving at a newly configured model while documenting each change made and its effect on the model. Another important recommendation regarding the research track process is that the calculation and presentation of standard diagnostics should be required to allow analysts, reviewers, and other stakeholders to best evaluate the model.

Age-structured models fitting relative biomass indices rely on the catch to be assumed to be known accurately to estimate population biomass. This assessment specified high coefficients of variation (CVs) on the removals to capture uncertainty in the landings. Although specifying this uncertainty is important, models like Stock Synthesis are designed to assume that the landings are known and with fairly high precision. Specifying high CVs on the removals provides flexibility to the model and causes Stock Synthesis to not fit the landings precisely, which can undermine the basis of these types of models. To handle uncertainty in the removals, we recommend developing several different sensitivity analysis scenarios (potentially a high, medium and low catch scenario) to test how different catch quantities could vary the model results. For each sensitivity analysis, maintain low CVs for the catch time series (see Francis 9 SEDAR 74 SAR SECTION V RESEARCH RECOMMENDATIONS January 2024 Gulf of Mexico Red Snapper 2017: Revisiting data weighting in fisheries stock assessment models, Fisheries Research, 192: 5-15).

All of the abundance indices were provided the same mean CV when they were input into Stock Synthesis. Assigning all indices to have the same CV does not consider the actual uncertainty behind each estimated annual index value and also informs Stock Synthesis that all indices are equally precise, which is not true. We recommend initially applying the CV from each annual estimate of the index, then applying a reweighting procedure to the indices to ensure the input variance assumptions are consistent with the output variance results. In addition, consider removing fishery dependent catch per unit effort indices given the number and quality of fishery independent indices available for this species. Incorporating fishery dependent indices could be a sensitivity analysis. It is important to acknowledge that for many species assessed in the Gulf of Mexico, we do not have enough or adequate fishery independent indices, and thus need to rely on including fishery dependent indices into these stock assessments. However, for red snapper, there are a number of fishery independent indices available that probably represent a more reliable index of abundance than fishery-dependent data sources.

The review panel found the stock assessment model to be overly complex, and discussed recommendations for how the model could be simplified. One of the ways we thought the model

could be simplified was by incorporating the discarded catch either directly into the landings or as its own fleet, instead of having the model try to fit to the discards. This would reduce the number of parameters that need to be estimated by eliminating retention functions for each time block associated with each fleet with discards. To accomplish this, external to Stock Synthesis, if needed, we recommend applying the discard mortality rate to determine the total number of dead discards. Given that there is uncertainty and potential bias in discarded catch, several different scenarios developed using the CVs of the discards (perhaps a low, medium and high scenario) could be sensitivity tested. If the dead discards are combined with landings to make up total removals prior to input into SS3, combine the discard length frequency data with the length frequency data from the landed catch; properly scale the discard composition data and the composition data from the landed catch. If incorporating discards as a fleet, then pair properly scaled discard composition data with the discard fleet.

Another way that the review panel thought the complexity of the model could be simplified was to reconsider the spatial structure of the model. The review panel recognizes that Gulf of Mexico red snapper are one stock from a biological, assessment, and management perspective. The stock identification group identified three potential sub-populations and recommended a three-area spatial structure be applied to the stock assessment model. There may be compelling reasons to model the stock in multiple areas, such as when there are differences in biology or population genetics, or differences in fisher behavior or fleet structure. The review panel acknowledges that there are some differences in fishing behavior and species targeting that differentiate the West Florida Shelf from the Panhandle of Florida and waters west of the Mississippi River.

The review panel recognizes that an important part of the SEDAR process is the development of consensus decisions by the stakeholders involved regarding the analysis and incorporation of data streams into the assessment, and the structure of the assessment model. This is a very useful way to ensure that important components of the system are captured in the data analysis and the assessment. However, the assessment scientists are those who are most knowledgeable about the implications of such decisions on the outcome of the Stock Synthesis model. As such, we recommend that the final decision on model spatial structure be made by the assessment analysts, given the information provided by the stakeholders together with their knowledge of available data, their respective sample sizes in each area, and fishing behavior. In this assessment, there were a limited number of data samples in the central region. As such, despite possible differences in fisher behavior between that region and the West Florida Shelf, the review panel recommended that the eastern and central region not be modeled separately.

In addition, the review panel found that the spatial delineation between the central region and the West Florida Shelf incorporated too much of the directed grouper fishing grounds into the central region such that the boundary would need to be reconsidered if a three-area model were to be pursued. Ultimately, the review panel recommended returning to a two-area model configuration. As the red snapper stock continues to rebuild, especially in areas east of the Mississippi River, more data on red snapper catch and composition will be available potentially making it possible to support a three-area model in the future. The review panel also recommended that a three area model could be developed as a sensitivity analysis. However, the review panel also recognizes that requesting multiple spatial partitions places an additional workload on the data preparation

team in the SEFSC and may not be feasible given current work assignments, until such processes are more automated.

Data from the GRSC were included in the stock assessment model as a one-year index of absolute abundance, by assigning it a catchability coefficient of 1 in the abundance index section of Stock Synthesis. After reviewing the Great Red Snapper Count documentation, the review panel (with the abstention of Dr. Sean Powers, one of the study PIs) determined that there were potential biases that had not been quantified and needed to be considered prior to including this information into the stock assessment. As discussed in an earlier section of this report, there is potential bias in the estimates of absolute abundance. As described, the average density estimated from the ROV sampling was nine times higher than the density estimated from acoustics at the same sampling stations. Due to differences like these, it is unlikely that the catchability (q) is equal to 1.

The review panel recognizes the GRSC data as a valuable source of information, regardless of whether it is used directly in the stock assessment. To consider including this data directly into the stock assessment model, the review panel recommends that the biases for the methods used in the GRSC be quantified and used to develop informed priors for the catchability of each sampling method, and an overall catchability prior for the survey as a whole. The review panel recommends that the calculated catchability prior then be used in the Stock Synthesis model to inform the estimation of catchability (q) for the GRSC estimates. This will allow the GRSC estimates to be most appropriately included in the Stock Synthesis model. In addition, assuming the selectivity for the survey is uniform across ages two and older is not appropriate. The review panel recommends incorporating the associated GRSC length frequency data into the assessment (after properly scaling the data), which could then help inform an associated estimated selectivity function.

The review panel agreed that fixing the steepness (h) of the stock-recruitment relationship at 0.99 does not properly reflect the life history characteristics of this species; doing so essentially generates the same average recruitment regardless of almost any stock size. Under this model configuration, the recruitment deviations were the only influence varying recruitment levels from one year to the next. If steepness cannot be estimated, or when allowed to be estimated, produces estimates at or close 0.99, then the review panel recommends developing several sensitivity analyses that test different steepness values (fixing the parameter) within an appropriate range for a species with a life history like that of Gulf of Mexico red snapper. In the configuration of Stock Synthesis we reviewed, the panel concluded that not requiring the recruitment deviations to average to zero was inappropriate because doing so undermines the integrity of the R_0 and B_0 parameters. In the model we reviewed, the recruitment deviations had a positive, upwards trend over time, giving the model the perception that a regime shift was taking place. As a result, the review panel recommends that the option to have the recruitment deviations sum to zero be used.

Several other recommendations from the review panel include smoothing any obvious and known outliers in the input data. The review panel recommends that the age of the population plus group be increased to an older age where there are less than a small percent of fish remaining in the virgin population. We also recommend testing the behavior of the assumed dirichlet distribution

for sensitivity of the parameter estimates to sample size starting values. Finally, to develop the sensitivity analysis that the review panel identified, we recommend that a matrix of possible and plausible combinations of parameters and processes that the analysts wish to test be developed to ensure a balance between testing the most important aspects of uncertainty while maintaining an appropriate workload.

